

Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

(I know what AI-generated code looks like, don't even think about it :>)

Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`

```
ryan@WIN-RO2UQQ0R3FN MINGW64 ~/source/repos/IT314AB_QUINO (feature/widgets)  
$
```

Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

`Text(`

`'Chiara Marie C. Canque\n'`

`'BSIT - 3\n'`

```
'My First Flutter Application\n'\n'August 11, 2026',\n)
```

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

Text → Application Title

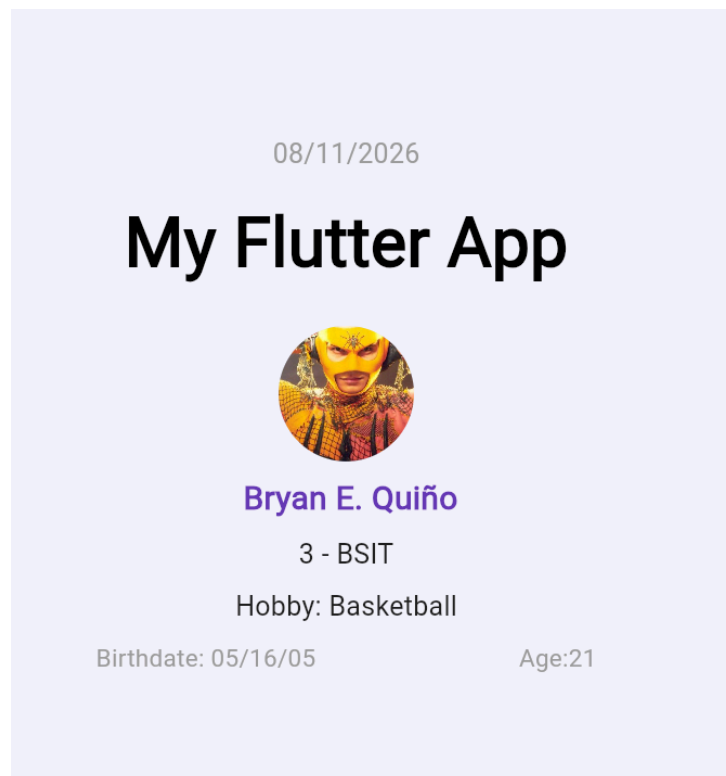
Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets
- Your application

```
    SizedBox(height: 10),
    Text(
      'Bryan E. Quiño',
      style: TextStyle(
        fontSize: 24,
        fontWeight: FontWeight.bold,
        color: Colors.deepPurple,
        fontFamily: 'serif',
      ),
    ),
    SizedBox(height: 10),
    Text(
      '3 - BSIT',
      style: TextStyle(
        fontSize: 20,
        color: Colors.black87,
        fontFamily: 'serif',
      ),
    ),
    SizedBox(height: 10),
    Text(
      '08/05/2026',
      style: TextStyle(
        fontSize: 18,
        color: Colors.grey,
        fontFamily: 'serif',
      ),
    ),
  ),
),
```



🚩 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

Course & Section

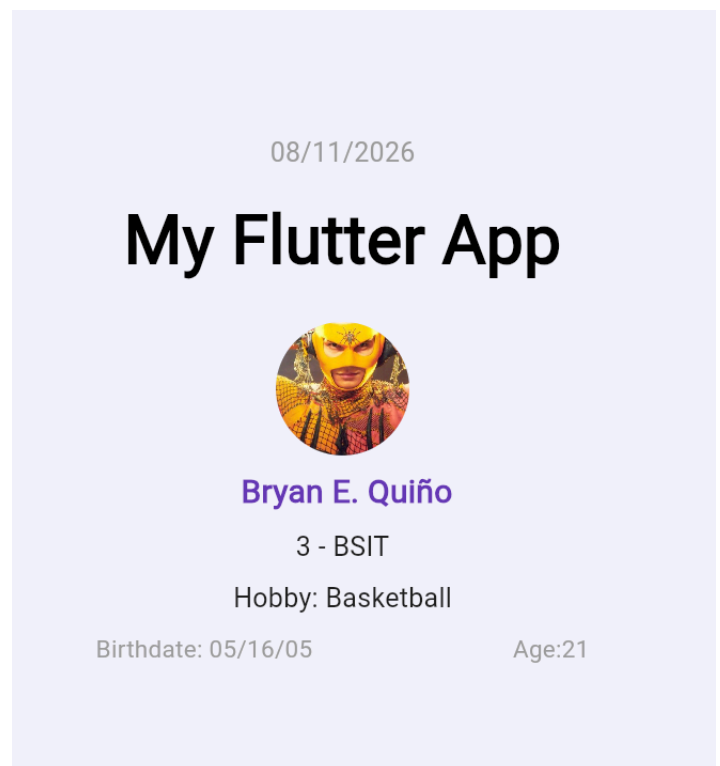
Application Title

Date

Screenshot:

- The `Column` in your code
- The four separate pieces of information displayed vertically

```
child: Column(  
  mainAxisAlignment: MainAxisAlignment.min,  
  children: const [  
    CircleAvatar(  
      radius: 50,  
      backgroundImage: AssetImage('lib/assets/gagamboy.jpg'),  
    ),  
    SizedBox(height: 10),  
    Text(  
      'Bryan E. Quiño',  
      style: TextStyle(  
        fontSize: 24,  
        fontWeight: FontWeight.bold,  
        color: Colors.deepPurple,  
        fontFamily: 'serif',  
      ),  
    ),  
    SizedBox(height: 10),  
    Text(  
      '3 - BSIT',  
      style: TextStyle(  
        fontSize: 20,  
        color: Colors.black87,  
        fontFamily: 'serif',  
      ),  
    ),  
    SizedBox(height: 10),  
    Text(  
      '08/05/2026',  
      style: TextStyle(  
        fontSize: 18,  
        color: Colors.grey,  
        fontFamily: 'serif',  
      ),  
    ),  
  ],  
)
```



🚩 Flag 3 – Add Another Element

Objective: Add another individual widget to your `Column`.

Add something that belongs to your profile.

Examples:

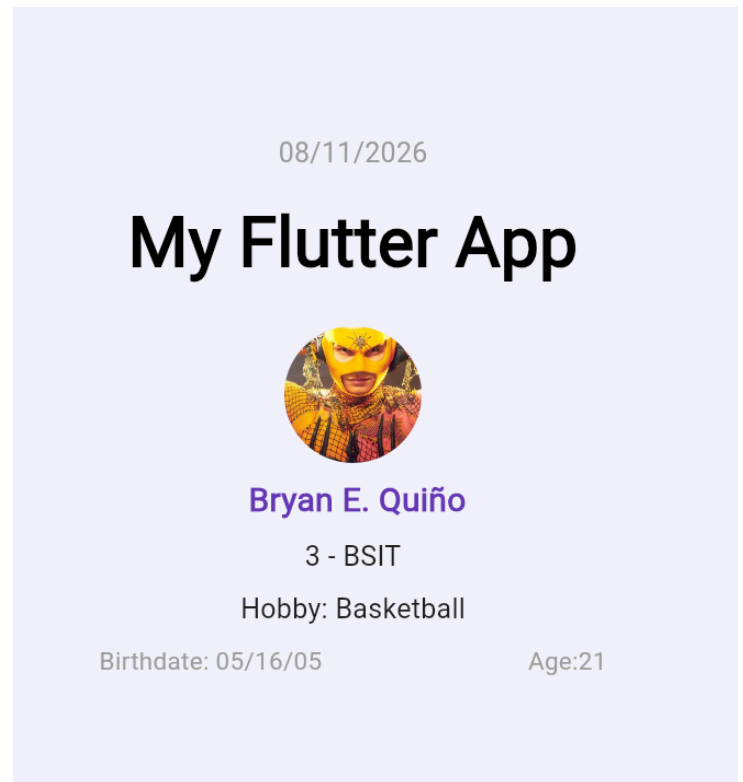
Hobby

It must be its **own widget**, not additional text inside another `Text`.

Screenshot:

- New widget in your `Column`
- Updated application

```
child: Column(  
  mainAxisAlignment: MainAxisAlignment.min,  
  children: const [  
    CircleAvatar(  
      radius: 50,  
      backgroundImage: AssetImage('lib/assets/gagamboy.jpg'),  
    ),  
    SizedBox(height: 10),  
    Text(  
      'Bryan E. Quiño',  
      style: TextStyle(  
        fontSize: 24,  
        fontWeight: FontWeight.bold,  
        color: Colors.deepPurple,  
        fontFamily: 'serif',  
      ),  
    ),  
    SizedBox(height: 10),  
    Text(  
      '3 - BSIT',  
      style: TextStyle(  
        fontSize: 20,  
        color: Colors.black87,  
        fontFamily: 'serif',  
      ),  
    ),  
    SizedBox(height: 10),  
    Text(  
      '08/05/2026',  
      style: TextStyle(  
        fontSize: 18,  
        color: Colors.grey,  
        fontFamily: 'serif',  
      ),  
    ),  
  ],  
)
```



Flag 4 – Put Two Things Side-by-Side

Objective: Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

`Column`

`|— Text`

`|— Text`

`|— Text`

`|— Text`

`|— Text`

`└— Row`

`|— Text`

`└— Text`

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side

```

Row(
  mainAxisAlignment: MainAxisAlignment.center,
  children:[
    Text(
      'Birthdate: 05/16/05' ,
      style: TextStyle(
        fontSize: 18,
        color: Colors.grey,
        fontFamily: 'serif',
      ), // TextStyle
    ), // Text
    SizedBox(width: 150),
    Text(
      'Age:21',
      style: TextStyle(
        fontSize: 18,
        color: Colors.grey,
        fontFamily: 'serif',
      ) // TextStyle
    ) // Text
  ], // Row
)

```

Birthdate: 05/16/05

Age:21

Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`SizedBox()`

Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet.**

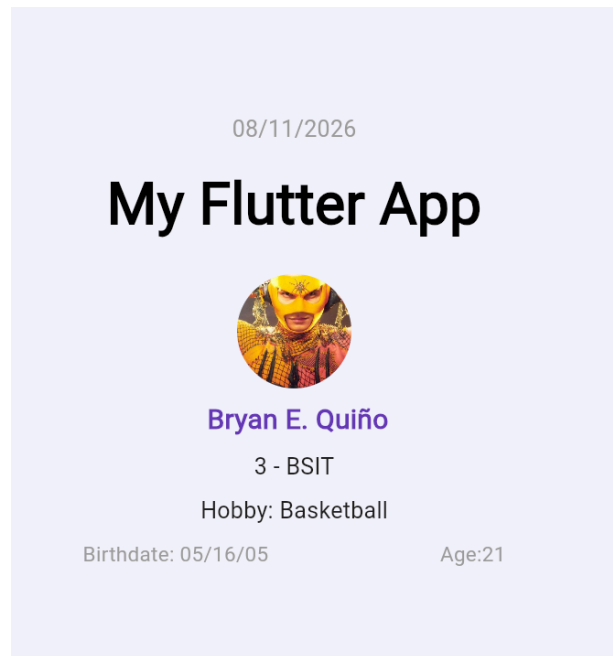
Screenshot:

- `SizedBox` usage
- Updated application

```

    SizedBox(height: 10),
    Text(
      'Bryan E. Quiño',
      style: TextStyle(
        fontSize: 24,
        fontWeight: FontWeight.bold,
        color: Colors.deepPurple,
        fontFamily: 'serif',
      ), // TextStyle
    ), // Text
    SizedBox(height: 10),
    Text(
      '3 - BSIT',
      style: TextStyle(
        fontSize: 20,
        color: Colors.black87,
        fontFamily: 'serif',
      ), // TextStyle
    ), // Text
    SizedBox(height: 10),
    Text(
      'Hobby: Basketball',
      style: TextStyle(
        fontSize: 20,
        color: Colors.black87,
        fontFamily: 'serif',
      ), // TextStyle
    ), // Text
    SizedBox(height: 10),

```



Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

Apply different styling to at least **three** elements.

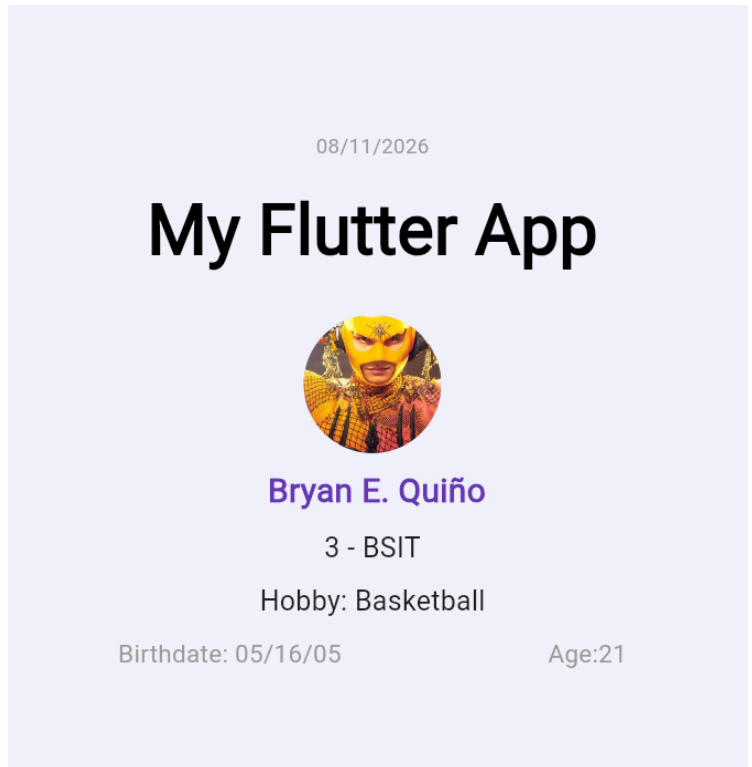
For example:

- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes
- Updated application

```
Text(
  '08/11/2026',
  style: TextStyle(
    fontSize: 15,
    color: Colors.grey,
    fontFamily: 'serif'
  ), // TextStyle
), // Text
SizeBox(height: 15),
Text(
  'My Flutter App',
  style: TextStyle(
    fontSize: 50,
    fontWeight: FontWeight.bold,
    color: Color.fromARGB(255, 0, 0, 0),
    fontFamily: 'Arial'
  ), // TextStyle
), // Text
SizeBox(height: 25),
CircleAvatar(
  radius: 50,
  backgroundImage: AssetImage('lib/assets/gaganboy.jpg'),
), // CircleAvatar
SizeBox(height: 10),
Text(
  'Bryan E. Quiño',
  style: TextStyle(
    fontSize: 24,
    fontWeight: FontWeight.bold,
    color: Colors.deepPurple,
    fontFamily: 'serif',
  ), // TextStyle
), // Text
SizeBox(height: 10),
Text(
  '3 - BSIT',
  style: TextStyle(
    fontSize: 20,
    color: Colors.black87,
    fontFamily: 'serif',
  ), // TextStyle
), // Text
SizeBox(height: 10),
Text(
  'Hobby: Basketball',
  style: TextStyle(
    fontSize: 20,
    color: Colors.black87,
    fontFamily: 'serif',
  ), // TextStyle
), // Text
SizeBox(height: 10),
```



🚩 Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

`mainAxisAlignment`

`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

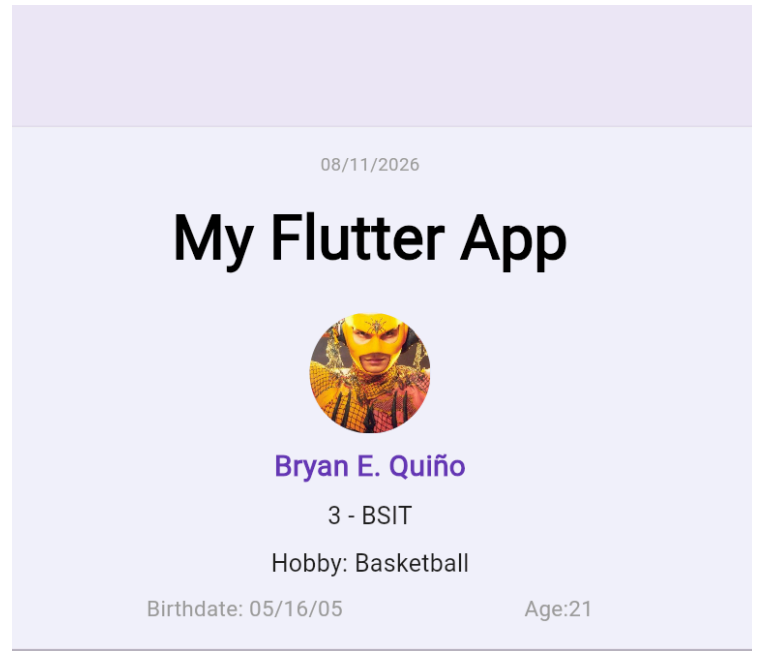
Screenshot:

- The properties you changed
- Updated application


```

crossAxisAlignment: CrossAxisAlignment.center,
children: const [
  Text(
    '08/11/2026',
    style: TextStyle(
      fontSize: 15,
      color: Colors.grey,
      fontFamily: 'serif'
    ), // TextStyle
  ), // Text
  SizedBox(height: 15),
  Text(
    'My Flutter App',
    style: TextStyle(
      fontSize: 50,
      fontWeight: FontWeight.bold,
      color: Color.fromARGB(255, 0, 0, 0),
      fontFamily: 'Arial'
    ), // TextStyle
  ), // Text
  SizedBox(height: 25),
  CircleAvatar(
    radius: 50,
    backgroundImage: AssetImage('lib/assets/gaganboy.jpg'),
  ), // CircleAvatar
  SizedBox(height: 10),
  Text(
    'Bryan E. Quiño',
    style: TextStyle(
      fontSize: 24,
      fontWeight: FontWeight.bold,
      color: Colors.deepPurple,
      fontFamily: 'serif',
    ), // TextStyle
  ), // Text
  SizedBox(height: 10),
  Text(
    '3 - BSIT',
    style: TextStyle(
      fontSize: 20,
      color: Colors.black87,
      fontFamily: 'serif',
    ), // TextStyle
  ), // Text
  SizedBox(height: 10),
  Text(
    'Hobby: Basketball',
    style: TextStyle(
      fontSize: 20,
      color: Colors.black87,
      fontFamily: 'serif',
    ), // TextStyle
  ), // Text
  SizedBox(height: 10),
  Row(
    mainAxisAlignment: MainAxisAlignment.center,

```



🚩 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second **Card** containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

Favorite Movie

The second card should also use:

- Individual **Text** widgets
- A **Column**
- At least one **Row**

- Spacing

Screenshot:

- Code for the second card
- Both cards displayed

```
// SECOND CARD
Card(
  elevation: 5,

  shape: RoundedRectangleBorder(
    borderRadius: BorderRadius.circular(15),
  ), // RoundedRectangleBorder

  child: Padding(
    padding: const EdgeInsets.all(20),

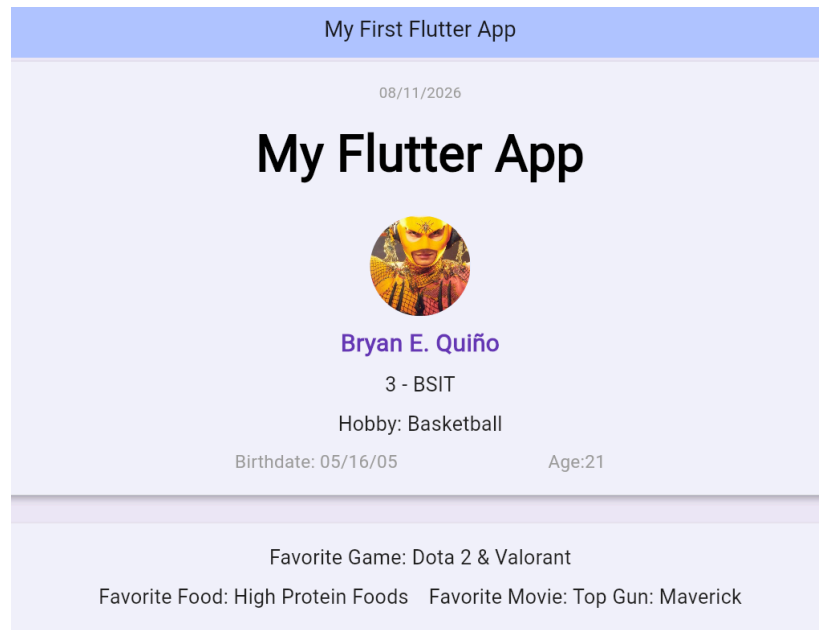
    child: Column(
      children: const [
        Text(
          'Favorite Game: Dota 2 & Valorant',
          style: TextStyle(
            fontSize: 20,
            color: Colors.black87,
            fontFamily: 'serif',
          ), // TextStyle
        ), // Text

        SizedBox(height: 10),

        Row(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text(
              'Favorite Food: High Protein Foods',
              style: TextStyle(
                fontSize: 20,
                color: Colors.black87,
                fontFamily: 'serif',
              ), // TextStyle
            ), // Text

            SizedBox(width: 20),

            Text(
              'Favorite Movie: Top Gun: Maverick',
              style: TextStyle(
                fontSize: 20,
                color: Colors.black87,
                fontFamily: 'serif',
              ), // TextStyle
            ), // Text
          ],
        ),
      ],
    ),
  ),
)
```



🚩 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card



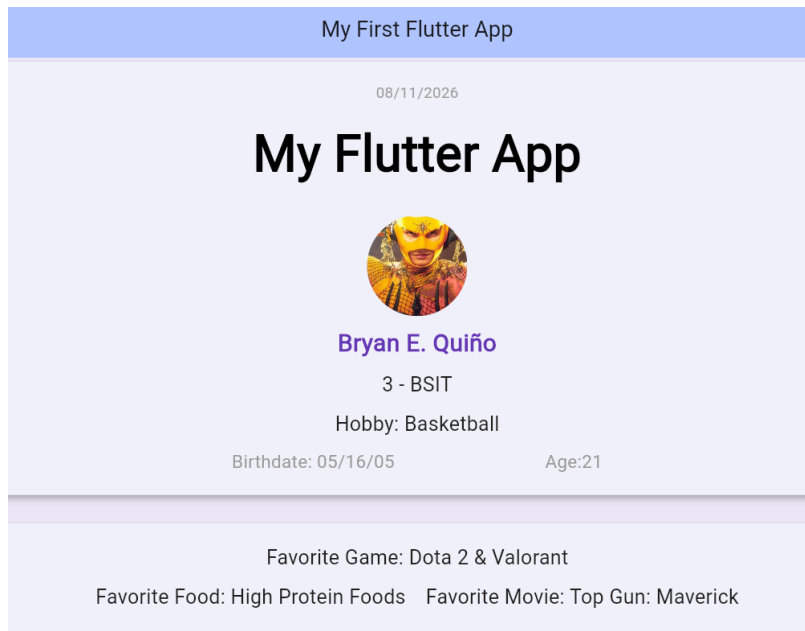
Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards
- Both cards visible in the application



🚩 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

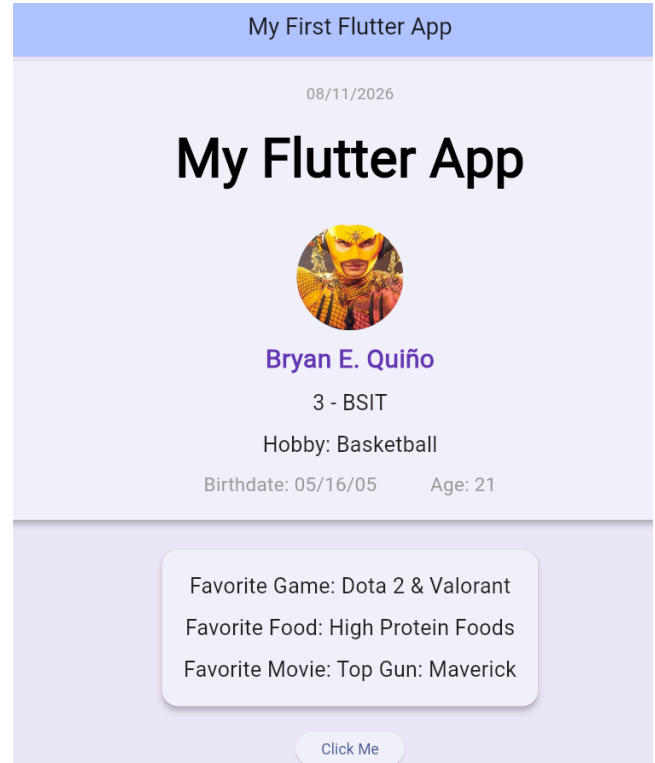
- `Icon`
- `Divider`
- `CircleAvatar`
- `Chip`
- `ElevatedButton`

Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code
- The widget visible in the application

```
// ELEVATED BUTTON
ElevatedButton(
  onPressed: () {
    print('Button pressed!');
  },
  child: const Text('Click Me'),
), // ElevatedButton
```



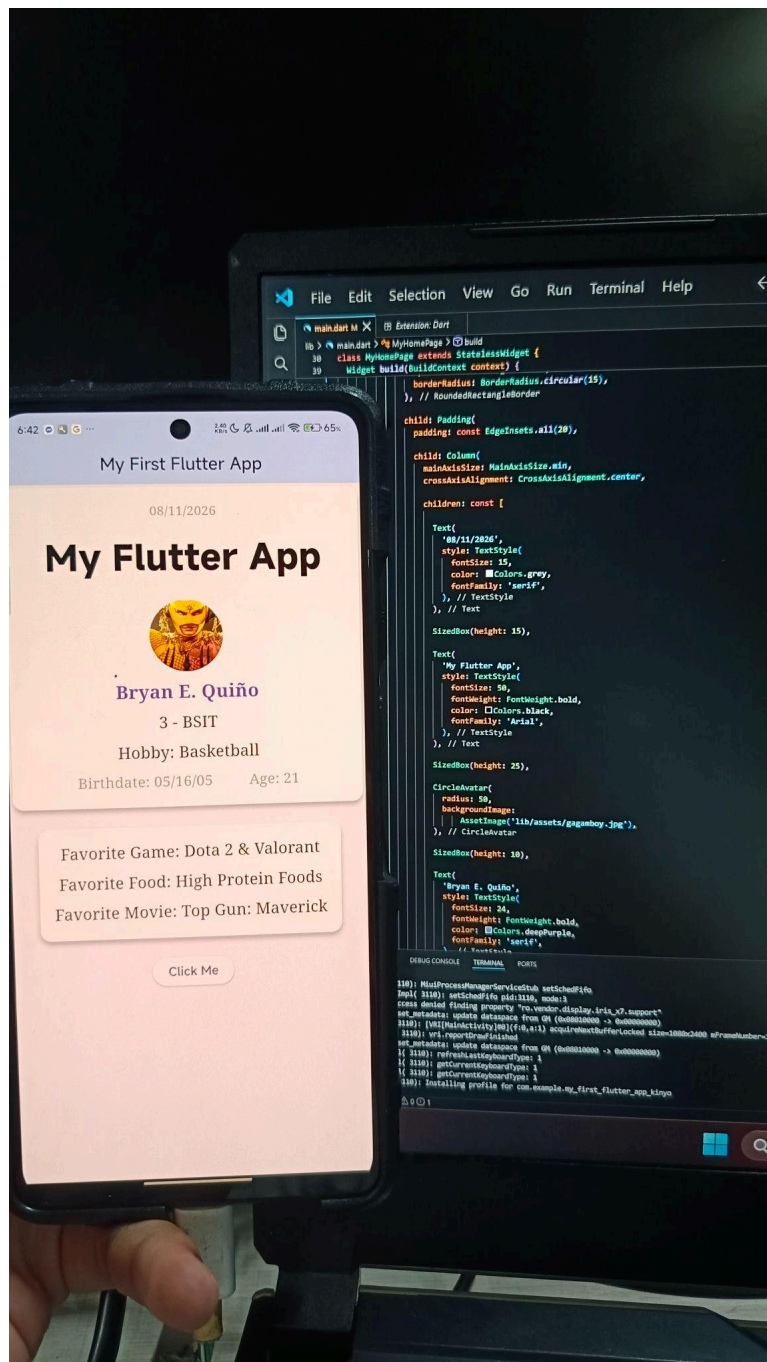
🚩 Flag 11 – Real Phone Test 📱

Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)



🚩 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit
- `feature/widgets` branch

